effect of deep sclerectomy is independent of external filtration (subconjunctival bleb), rarely occurring in only 5% of patients, and cite an article on viscocanalostomy as a reference.[2] Again, we feel it necessary to distinguish viscocanalostomy from deep sclerectomy with or without an implant. Although both are nonpenetrating surgeries, bleb formation after viscocanalostomy is rare due to tight closure of the superficial flap[2] (which was not performed by the authors in their viscocanalostomies in this report), whereas low, diffuse blebs are commonly found in deep sclerectomy, where the superficial scleral sutures are loosely tied.[4]

The authors should be commended for attempting a technically demanding procedure in these very challenging cases. However, it must be stated that this study was not able to appropriately measure the effectiveness of deep sclerectomy in refractory congenital glaucoma, the stated objective of the article. This study can only conclude that performing deep sclerectomy is technically difficult in these cases. It may be a fact that the authors had the intention of doing deep sclerectomy, but as the saying goes, some roads are paved with good intentions.

We do feel that the role of nonpenetrating glaucoma surgery in congenital glaucoma is yet undefined. In our experience, we have found reasonable success with an improved safety profile using this approach, and we will be publishing our results soon.

Tarek Shaarawy, MD
Giza, Egypt

Iqbal Ike K. Ahmed, MD, FRCS(C)
Toronto, Canada

References

1. Lüke C, Dietlein T, Jacobi PC, et al. Risk profile of deep sclerectomy for treatment of refractory congenital glaucomas. Ophthalmology 2002;109:1066–71.
2. Stegmann R, Pienaar A, Miller D. Viscocanalostomy for open-angle glaucoma in black African patients. J Cataract Refract Surg 1999;25:316–22.
3. Tixier J, Dureau P, Becquet F, Dufier JL. Deep sclerectomy in congenital glaucoma: preliminary results [in French]. J Fr Ophthalmol 1999;22:545–8.
4. Shaarawy T, Karlen M, Schnyder C, et al. Five-year results of deep sclerectomy with collagen implant. J Cataract Refract Surg 2001;27:1770–8.

Author reply

Dear Editor:
In their comment on our retrospective case series, Drs. Shaarawy and Ahmed state that our results contradict those of Tixier et al,[1] who published reasonable success rates after deep sclerectomy in patients with congenital glaucoma. In this context they may have overlooked that in our series all 10 eyes had a history of antiglaucomatous surgery with an average of 2.5 glaucoma interventions (Table 1), as indicated by the term "refractory congenital glaucoma." In the cited series of Tixier et al, most of the eyes had no previous surgery, with only 3 of 12 eyes having two interventions before deep sclerectomy. Thus, this comparison remains at least questionable.

Drs. Shaarawy and Ahmed recommended carefully distinguishing viscocanalostomy from deep sclerectomy due to different bleb rates. They indicated that bleb formation after viscocanalostomy is rare,[2] whereas it is commonly found after deep sclerectomy.[3] However, this substantially contradicts the recent review article[4] from their own study group stating that "50% of patients who underwent a viscocanalostomy had a subconjunctival filtering bleb." We feel that a watertight closure of a thin scleral flap in a buphthalmic eye is more likely a theoretical aim than practically feasible.

Deep sclerectomy is the essential surgical step of nonpenetrating techniques including viscocanalostomy. The first goal of these techniques is to unroof Schlemm's canal by the dissection of the deep scleral flap and thereby achieve a significant aqueous flow into the intrascleral space. In our series of refractory congenital glaucomas, in 60% of eyes this goal was not reached. However, a considerable rate of complications was observed.

Based on these results, we feel justified in drawing attention to the possible risks of nonpenetrating surgery in these rare cases of refractory congenital glaucoma with extremely thin sclera and pathological topography of Schlemm's canal.

Christoph Lüke, MD
Thomas S. Dietlein, MD
Philipp C. Jacobi, MD
Walter Konen, MD
Günter K. Krieglstein, MD
Köln, Germany

References

1. Tixier J, Dureau P, Becquet F, Dufier JL. Deep sclerectomy in congenital glaucoma: preliminary results [in French]. J Fr Ophthalmol 1999;22:545–8.
2. Stegmann R, Pienaar A, Miller D. Viscocanalostomy for open-angle glaucoma in black African patients. J Cataract Refract Surg 1999;25:316–22.
3. Shaarawy T, Karlen M, Schnyder C, et al. Five-year results of deep sclerectomy with collagen implant. J Cataract Refract Surg 2001;27:1770–1778.
4. Mermoud A. Sinusotomy and deep sclerectomy. Eye 2000; 14(pt 38):531–5.

Dear Editor:
We have a comment on the article on the risks profile of deep sclerectomy for treatment of refractory congenital glaucomas by Luke et al.

The pressure-lowering effect of deep sclerectomy is independent of external filtration only when the superficial scleral flap is tightly secured, as described in Stegmann's viscocanalostomy. External filtration is also absent when the scarring process is so important that the pressure-lowering effect of the surgery is minimal. In the article by Luke et al, only one of the eyes that underwent deep sclerectomy alone, with or without perforations, developed a filtering bleb. It was, moreover, the only one that achieved intraocular pressure (IOP) control. This is the reason why, to optimize all the possible outflow mechanisms past the trabeculodescemetic membrane, we tie our superficial flap very loosely, provided no visible perforation is present.

Of the 10 eyes in this series, only 2 had uneventful pure

deep sclerectomy. In two eyes, percolation through the trabeculodescemetic membrane was weak, so the procedure was supplemented by a trabeculotomy. In another four eyes, identification of Schlemm's canal was not made, so no deep sclerectomy alone nor added trabeculotomy could be done, and the procedure was converted into a guarded filtering procedure. Finally, in two eyes an inadvertent visible perforation of the trabeculodescemetic membrane occurred. The report of a high percentage of conversion to perforating filtering surgery and potential associated postoperative complications differs from our own experience in refractory congenital glaucoma. After both pure deep sclerectomies, hyphemas were reported as complications, although trivial. These may further result from possible microperforations, with a resultant high and abrupt change in IOP. We agree that deep sclerectomy alone is not applicable in most cases with complicated refractory congenital glaucomas. We make small visible perforations of the trabeculodescemetic membrane with the tip of a 30-gauge needle. Our incidence of nonidentified Sclemm's canal in congenital refractory glaucomas, however, is low.

Considering that only pure nonpenetrating filtering surgery is reported in this article, the IOP control after 19 months of follow-up was 50%, which, considering the high-risk failure rate for refractory congenital glaucoma, is not so bad.

E. Ravinet, MD
A. Mermoud, MD
Lausanne, Switzerland

Author reply

Dear Editor:
Drs. Ravinet and Mermoud detailed their deep sclerectomy techniques for treatment of refractory congenital glaucomas in their comment on our retrospective case series.

Between our observation and the experience of Drs. Ravinet and Mermoud there is complete consensus that nonpenetrating deep sclerectomy alone is not applicable in most cases with refractory congenital glaucoma.

To enhance aqueous outflow after deep sclerectomy they needle the trabeculodescemetic membrane, which causes macroperforations. Additionally, to allow subconjunctival filtration, only loose readaptation of the superficial scleral flap is performed. In our opinion this technique is a variant of filtering surgery or, as we would term it, a "penetrating deep sclerectomy."

Christoph Lüke, MD
Thomas S. Dietlein, MD
Philipp C. Jacobi, MD
Walter Konen, MD
Günter K. Krieglstein, MD
Köln, Germany

Sildenafil-Associated NAION

Dear Editor:
Pomeranz and colleagues' case series describing the occurrence of nonarteritic anterior ischemic optic neuropathy (NAION) after ingestion of sildenafil in five subjects invites speculation about the causative mechanism.[1] Their thesis primarily emphasized the notions of disrupted autoregulation in the optic nerve head (ONH) or a toxic optic neuropathy from potentiation of nitric oxide by sildenafil. Importantly, although their five patients had crowded discs, mention of systemic arterial hypotension as a possibile etiology was omitted. Systemic hypotension is particularly relevant in the context of a structurally congested ONH that has vulnerable watershed zones in its microcirculation.[2]

In this regard, the manufacturer (Pfizer Pharmaceuticals, New York, NY) has reported that sildenafil reduces blood pressure (independent of the amount of drug) in healthy supine subjects, especially 1 to 2 hours after administration. Subsequent studies have confirmed that sildenafil can produce a small decrease in systolic and diastolic blood pressure, with some groups finding a fall as large as 20 mmHg (systolic) in certain individuals, suggesting perhaps a difference in interindividual sensitivity to the drug.[3,4] Some workers examining the hemodynamic effects of sildenafil have used conventional sphygmomanometry at the brachial artery to assess effects on blood pressure. It is rather thought provoking that sphygmomanometry may significantly underestimate hemodynamic disturbances (e.g., the fall in aortic systolic and pulse pressures) induced by vasodilator agents when compared with relatively sophisticated techniques.[5] Therefore, hypotension and other hemodynamic changes in the central circulation that affect ONH perfusion after vasodilator therapy may be overlooked when attention is focused on detecting changes in brachial arterial pressure with sphygmomanometry.[6] These transient fluxes in the central circulation (though not florid enough to produce acute symptomatic hypotension) may be sufficient to elicit the final insult of critical ischemia in the anatomically susceptible ONH.

The role of arterial hypotension deserves explicit consideration in these five patients, where crowded ONHs were exposed to a drug known to modify systemic hemodynamics. Moreover, an allusion to this established mechanism does not invalidate Pomeranz et al's proposition that sildenafil might interfere with regulation of ONH microvasculature, because both local and systemic circulatory effects could conspire in the pathogenesis. Their alternative theory of a toxic optic neuropathy from sildenafil is less compelling because all five patients have clinical and investigative evidence compatible with classic NAION.

It is difficult to be specific about the mechanisms responsible for NAION secondary to sildenafil because rigid statements cannot be supported by existing evidence. Certainly, we cannot be dogmatic or claim omniscience because the pathophysiology of NAION is still poorly characterized. Suffice it to say, there are many predisposing and precipitating factors for NAION, and various permutations of these risks can generate ischemia at the ONH.[7] Sildenafil (like other antihypertensives or, indeed, like other acquired risks) may serve as a trigger given a strong enough predisposition.

Jagdeep Singh Gandhi, MB, ChB
London, England